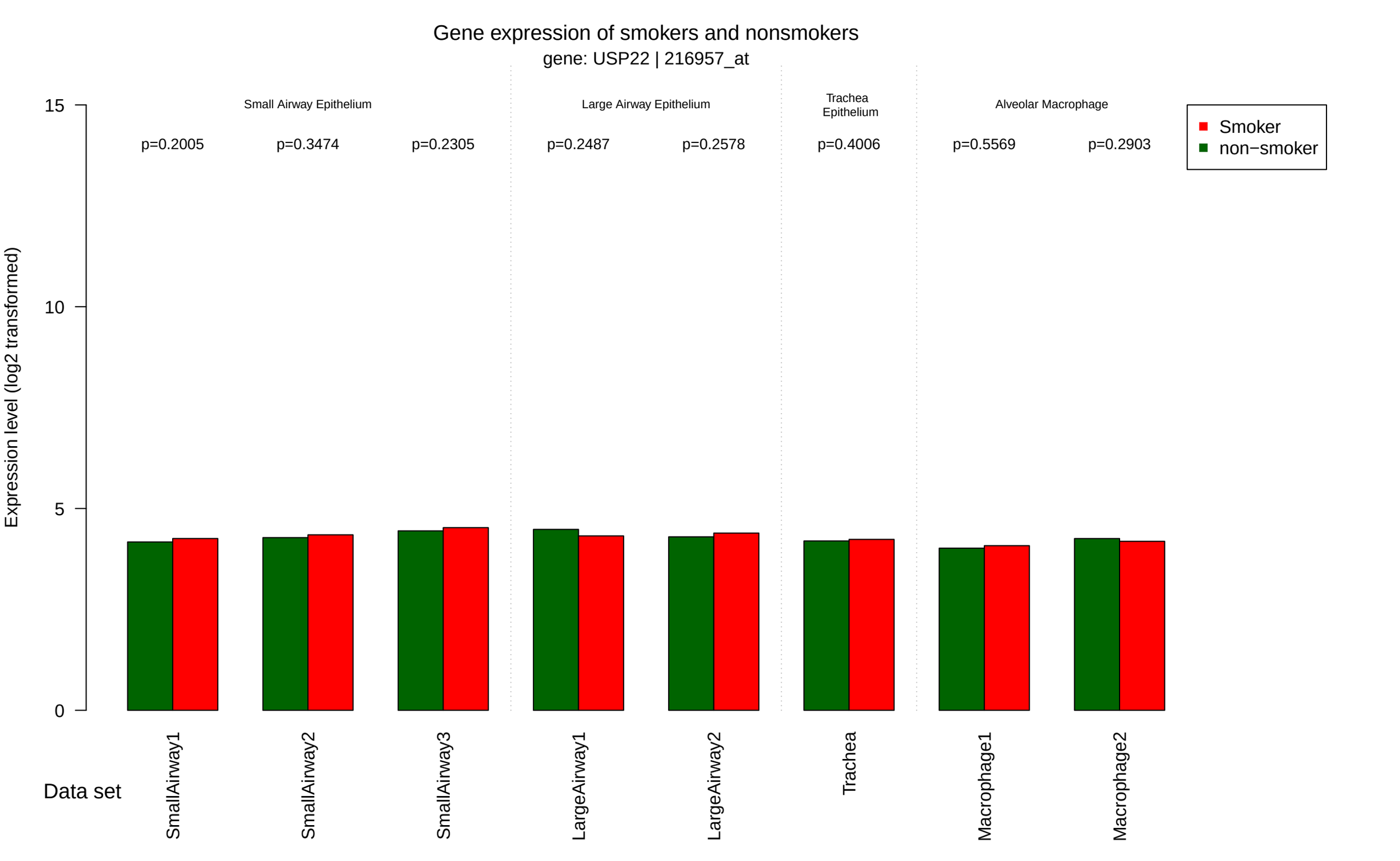
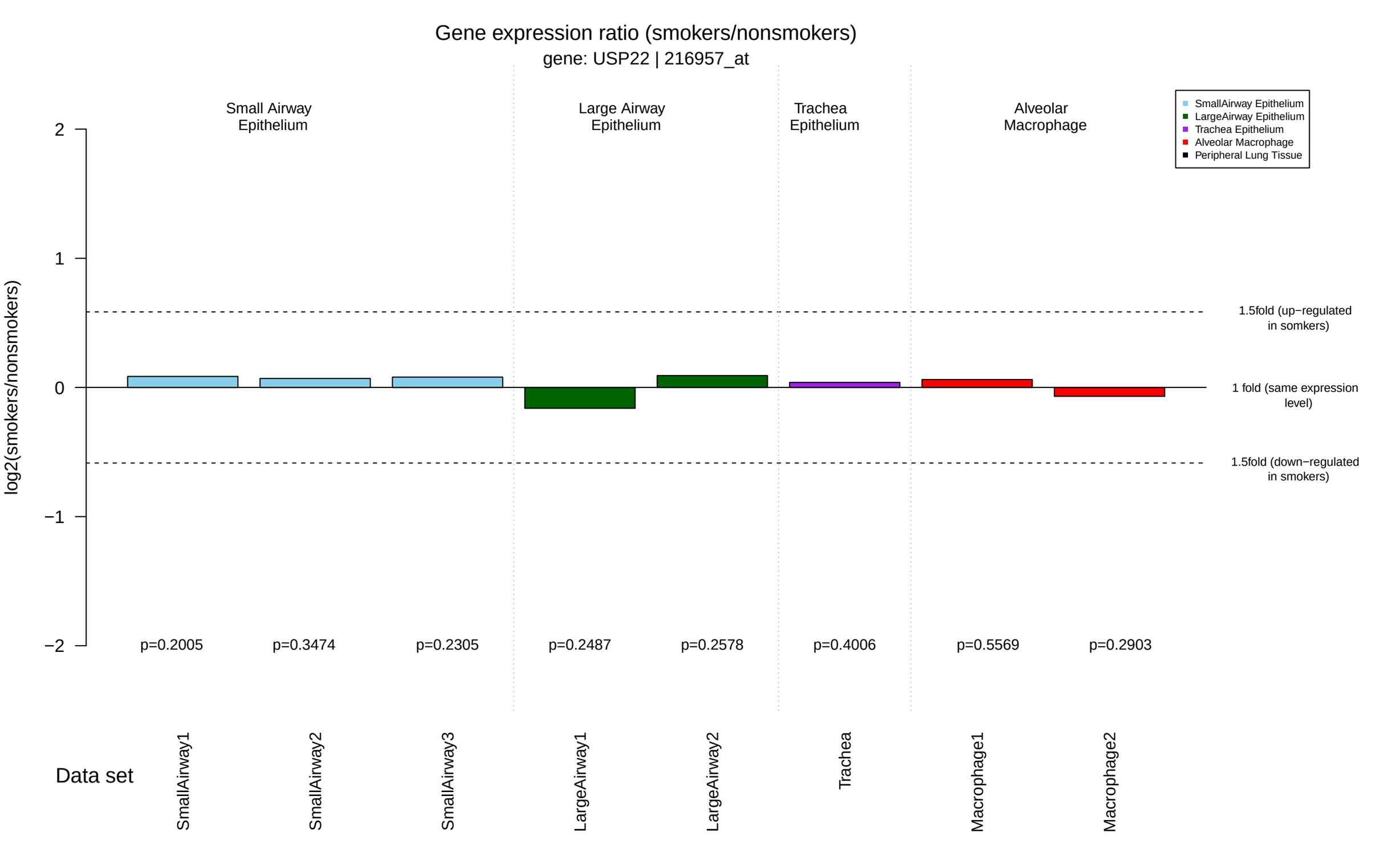
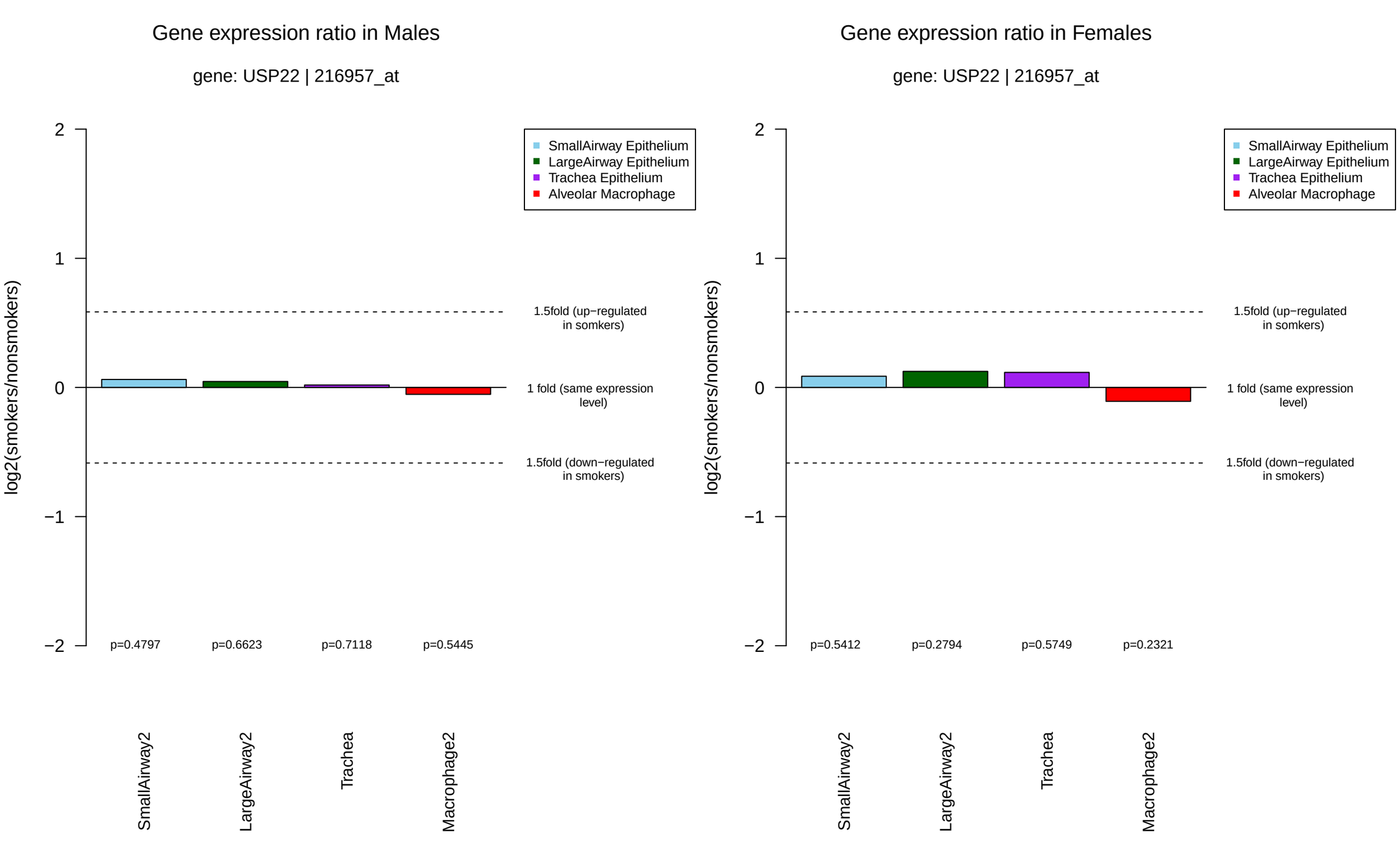
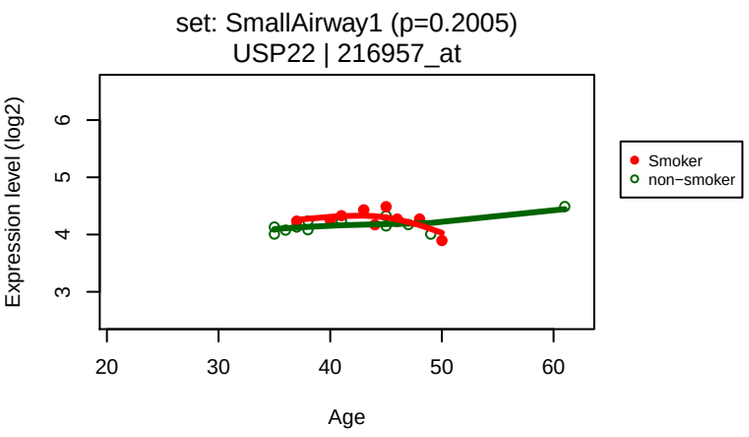
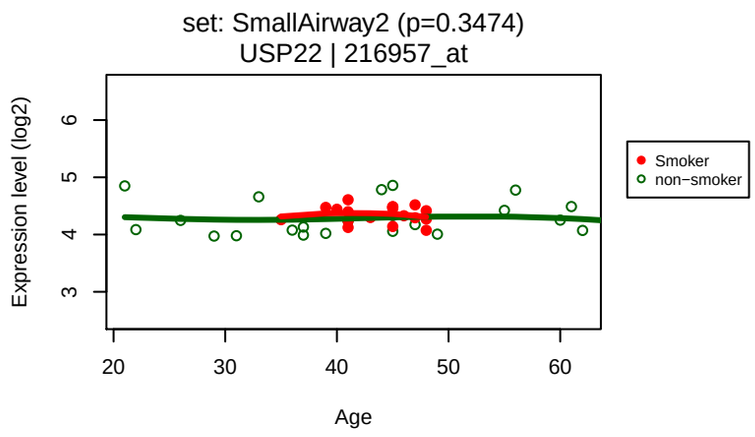
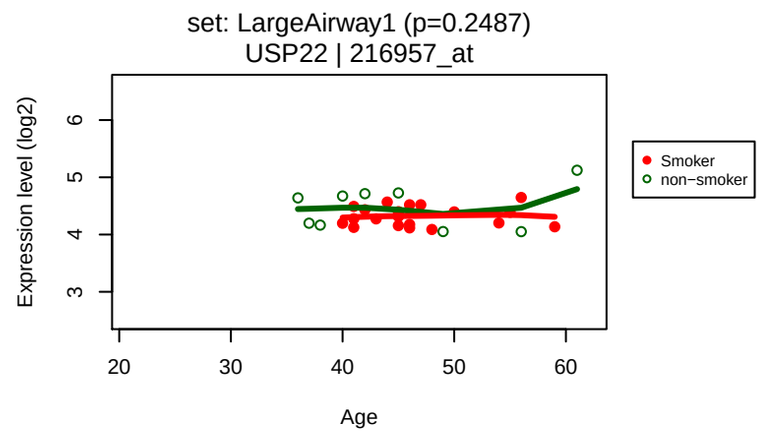

Gene expression of smokers and nonsmokers
gene: USP22 | 216957_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2005
p=0.3474
p=0.2305
p=0.2487
p=0.2578
p=0.4006
p=0.5569
p=0.2903
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

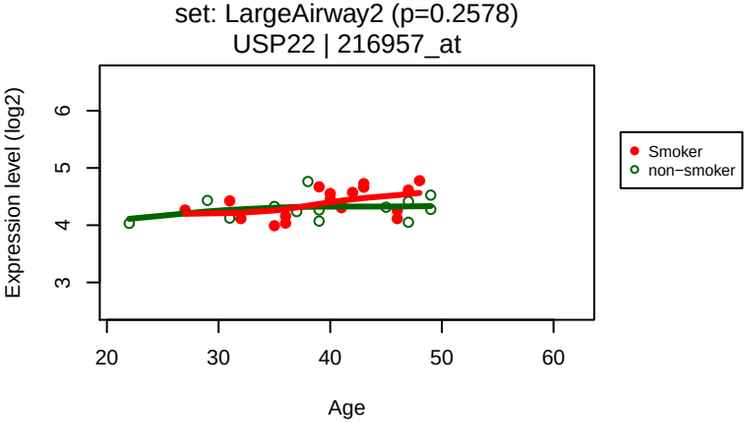

Gene expression ratio (smokers/nonsmokers)
gene: USP22 | 216957_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2005
p=0.3474
p=0.2305
p=0.2487
p=0.2578
p=0.4006
p=0.5569
p=0.2903
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

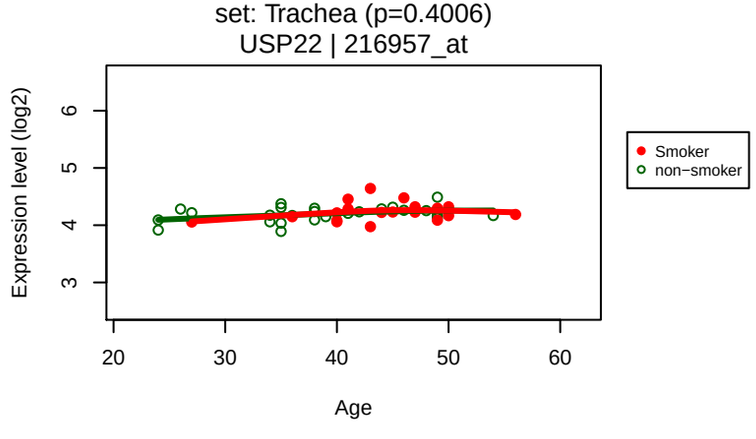

Gene expression ratio in Males
gene: USP22 | 216957_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4797
p=0.6623
p=0.7118
p=0.5445
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: USP22 | 216957_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5412
p=0.2794
p=0.5749
p=0.2321
SmallAirway2
LargeAirway2
Trachea
Macrophage2

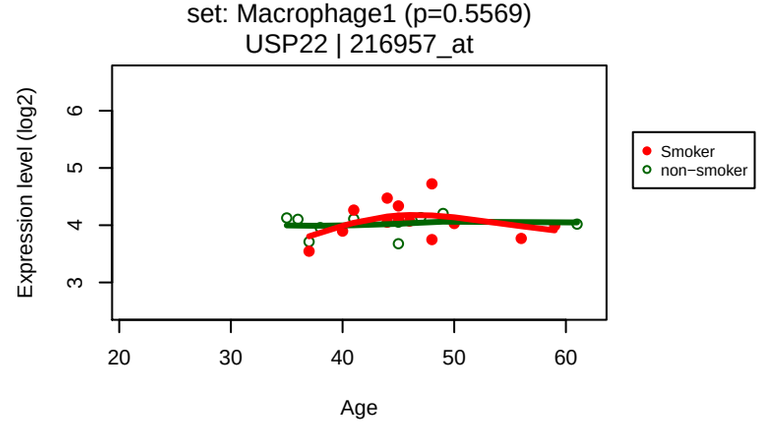

set: SmallAirway1 (p=0.2005)
USP22 | 216957_at
Expression level (log2)
Age
Smoker
non−smoker

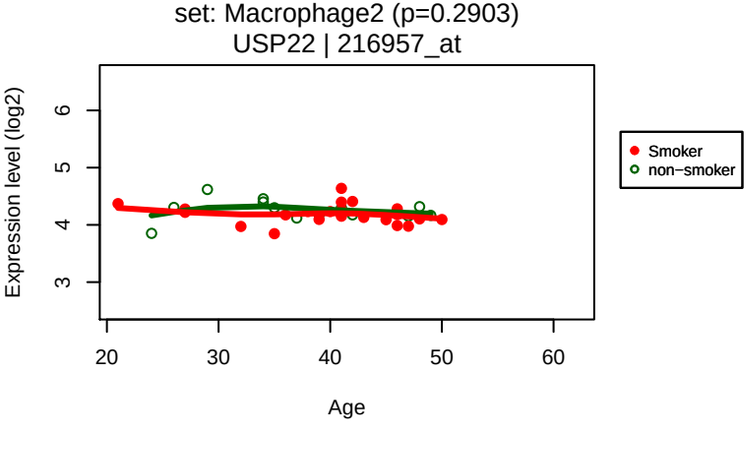

set: SmallAirway2 (p=0.3474)
USP22 | 216957_at
Expression level (log2)
Age
Smoker
non−smoker

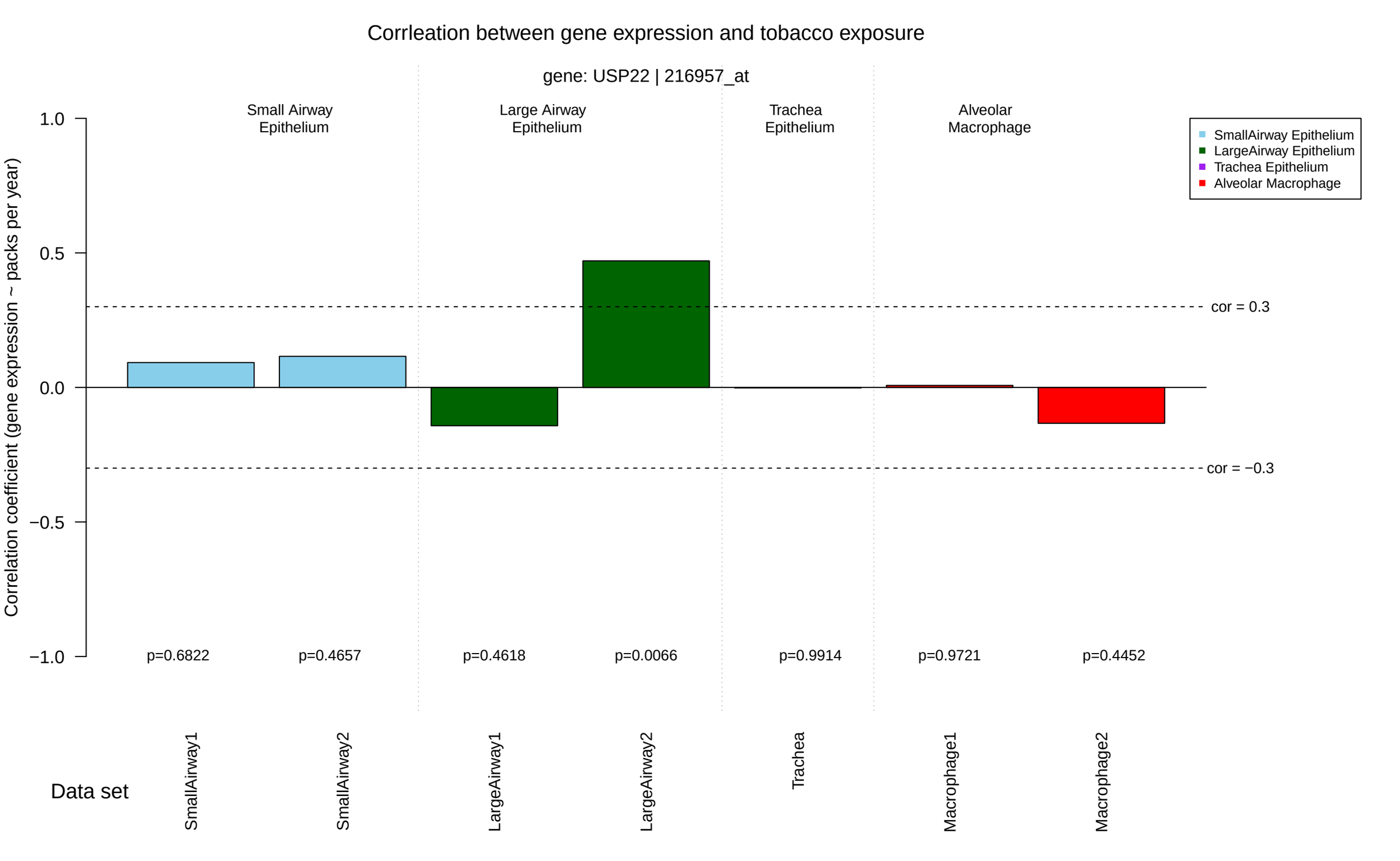

set: LargeAirway1 (p=0.2487)
USP22 | 216957_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2578)
USP22 | 216957_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4006)
USP22 | 216957_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5569)
USP22 | 216957_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2903)
USP22 | 216957_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: USP22 | 216957_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6822
p=0.4657
p=0.4618
p=0.0066
p=0.9914
p=0.9721
p=0.4452
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2